

Research on Motor Imagery Signal Recognition Method Based on Wavelet Packet Transform and ConvLSTM

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Abstract

Brain Computer Interface (BCI), as a novel method of information communication and control, is an interdisciplinary research topic which involves neuroscience, signal processing, and pattern recognition. The BCI system based on motion imagination is considered the most promising type of brain computer interface system. The key to the BCI system lies in the analysis and processing of EEG signals. This article combines the characteristics of CNN and LSTM models, and combines wavelet packet transform to propose a ConvLSTM based imaginative signal recognition model. The model first uses wavelet packet transform to extract features of EEG signals, then uses convolution operation for spatial features extraction of EEG signals, and uses LSTM network for temporal features extraction of EEG signals. ConvLSTM model is able to extract local high-level features and take the advantage of LSTM model to predict the sequence and the Wavelet Packet Transform has the advantage including multi-resolution analysis, spatial locality, simple and fast computation, and sparsity. The simulation experiment results demonstrate that the method is feasible.

CCS Concepts

• **Computing methodologies** → Machine learning; Machine learning approaches; Neural networks.

Keywords

BCI, Motor Imagery, ConvLSTM, Wavelet Packet Transform

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1 INTRODUCTION

With the development of cognitive science research and of signal processing technology, it has been discovered that there is a necessary connection between electroencephalogram (EEG) signals and

human thinking. Research has shown that bioelectric signals can be detected in corresponding brain regions simply through the process of imagining movement, the electrical signals will also undergo different changes with the limb movements of imagining movement and the Brain Interface System (BCI) which is motor imagery based was developed on the groundwork of this conclusion. In addition to the field of rehabilitation medicine, researchers also dedicate to applying BCI systems to military, entertainment, driverless cars, and environmental control fields, in which they have achieved important research attainments. And the ConvLSTM model used in this article is now widely adopted by researchers to recognize users' motion purposes. This model improves motion recognition accuracy and therefore promoting the application of BCI technology in daily life.

Brain motor imagery refers to the activity of repeatedly simulating and imagining certain physical movements without actual physical motions. When a certain area of the cerebral cortex is stimulated, the metabolic activity of that area is enhanced and the amplitude of α and β waves oscillation significantly decreases. This electrophysiological phenomenon is known as Event Related Desynchronization (ERD) [1]. On the other side, the amplitude of α and β waves oscillations increases when they are in a resting or suppressed state in certain areas of the brain, which is called Event Related Synchronization (ERS) [2]. Both actual and imagined limb movements in the human body can stimulate the corresponding sensory motor regions of the brain, arousing ERD/ERS phenomena in parallel frequency range. For the left and right hands motor imagery, there is a clear contralateral dominance in corresponding brain motor areas. When imagining the movements of left hand and right hand, the ERD/ERS phenomenon occurs in the μ rhythm (8-13Hz) and β rhythm (18-26Hz). When imagining foot movements, the ERD appears at 7-8Hz and 20-24Hz. When imagining tongue movements, the ERS phenomenon occurs at 10-11Hz. By accurately identifying these features, they can be transformed into specific instructions for use in the control of brain computer interface systems. The biggest advantage of a BCI system which is motor imagery based is that its control signal is an endogenous induced EEG, which does not require specific external stimuli during motor imagery. However, there more training experiments is still needed, meanwhile, individual differences yet exist, thus it is necessary to resolve frequency band of EEG signals to improve recognition accuracy [3].

BCI based on imagination is an important type of BCI that is not only able to control wheelchairs to improve mobility of patients who suffer from severe lower limb movement disorders, but also to train these patients to recover their motor function to a certain extent. Motor imagery is a process of cognition in which subjects

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imagine performing actions without actual performance. This is a dynamic state during which the representation of specific motor actions is internally stimulated without any output of motor. In the medical field, motor imagery is applied in rehabilitation training of patients, combined with BCI technology for brain controlling peripherals, and can be used as a substitute for residual limbs to help patients restore their self-care ability and mobility.

The key to BCI systems lies in the processing and analysis of EEG signals. In order to promote the development of BCI systems, researchers hope to obtain an EEG signal classification model with strong generalization capability and high classifying accuracy. Therefore, how to apply deep-learning-based methods to EEG signal analysis, capturing more recognizable and representative deep features in highly complex EEG image data is an extremely valuable research direction for future BCI research.

2 Related Work

The traditional method for analyzing EEG signals generally involves manual extraction of the EEG signal feature information, and then using machine learning methods to construct the mapping relationship between the feature information of EEG signal and motor imagery. Liu [4] uses Partial Least Squares Regression (PLS) as the feature selection algorithm. PLS not only provides a reasonable regression model in the process of linear regression, but also conducts Principal Component Analysis (PCA) and Canonical Correlation Analysis (CCA), achieving the effect of dimensionality reduction and the extraction of data samples with features gaining higher significance than that processed by PCA algorithm. Yang [5] proposes an EEG signal recognition algorithm based on Sparse Common Space pattern and Sparse Logistic Regression. This algorithm performs two sparse optimizations on the EEG signal recognition algorithms based on common space pattern and linear discriminant analysis. By utilizing the two sparsity during feature extracting and classifying, it adaptively preserves appropriate classification features and achieves adaptive selection of channels and features. Guo [6] proposes using three various methods of analyzing: time-domain analysis, frequency-domain analysis, and spatial analysis to analyze EEG signals, and then adopting the Common Space Pattern Method (CSP) and its derivative algorithm, the Filter Bank Common Space Pattern Method (FBCSP) for the feature extraction of the reconstructed EEG, introducing the wavelet packet transform to improve the FBCSP, and a feature extracting method based on the Wavelet Packet Common Space Pattern Method (WPCSP) was designed and, finally, a four-class classification recognition is performed on the four types of motion imagination EEG which has been feature extracted. The classifier used is an improved decision tree Support Vector Machine (SVM), which can be divided into four categories. Yang [7] proposes using a decreasing connection structure to improve the Focal loss function by introducing paradigms, transforming the model from empirical risk loss to structural risk loss. Wang [8] proposes a joint classification method based on wavelet transform feature extracting, Principal Component Analysis (PCA), and Support Vector Machine (SVM) for motor imagery EEG signals of left hand and right hand. Due to the high dimensionality of the extracted signal features, the dimensionality reducing algorithm of PCA and the SVM classification algorithm is adopted.

The use of training samples to train SVM, finding the weight vector and bias that have the best classification performance, and finally obtaining the decision function by which the dimensionality reduced feature vector is used in the decision function to obtain the classification result is carried out then. Bao [9] proposes to analyze EEG signals from the perspective of phase characteristics for non-stationary and non-Gaussian characteristics of EEG signals. By extracting the intrinsic mode function of EEG signals, he constructs a phase fluctuation function using the Hilbert-Huang transform. He uses detrended fluctuation analysis to perform scale analysis on EEG, extracting features of EEG signals, and using regularization discriminant analysis to identify EEG intentions. However, the feature extraction performed mainly relies on domain knowledge and subjective experience, which may lead to the loss of useful information in EEG signals, and subsequently affecting classifier performance.

Traditional machine learning based EEG signal analysis methods have achieved certain research results, but these methods also require a large amount of priori knowledge to search for data features, resulting in their performance not being fully applicable to real-time online BCI systems.

Deep learning methods can automatically extract features with rich representation capabilities, and can directly input raw EEG signal data into models of deep learning, without the need for feature extraction, to construct end-to-end deep learning models. Deep learning is now becoming a powerful tool for studying motor imagery behavior. Chang [10] proposes using causal networks for the recognizing multiple types of motor imagery EEG signals. Firstly, a multi variable Granger causality relationship is used to construct a motor imagery effect network. Then, a convolutional neural network is trained by the effect network matrix and the convolutional neural network model is determined. Finally, the model is used to classify multiple types of motor imagery EEG signals, achieving good accuracy. He [11] uses Simplified Convolutional Neural Network (SCNN) to classify EEG signals of motor imagery, and removed convolutional layers from standard CNN to optimize network structure and prevent effective feature loss. Then, two one-dimensional convolutional kernels are adopted to extract time-domain and frequency-domain features, respectively, to reduce network parameters and improve training speed. Zheng et al. [12] proposes a new overlapping time slice training strategy for deep learning, which addresses the strong time-varying nature of EEG signals, insufficient training samples, and high feature dimensions, etc. Basing on the existing time slice strategy (cropped), they adopt an overlapping time slice strategy (overlapped), and construct a new loss function calculation and label prediction method based on the overlapping time slice set, it can further increase training samples, reduce the feature space of individual samples, and thus improve the performance of deep neural networks in EEG signal recognition. Xu [13] proposes a Dual Head Dual Attention Networks (DHDANet) deep learning recognition model with a dual input and dual attention mechanism to address the issue of low classification accuracy in EEG motor imagery classification and recognition when the test object is not in the sample of the constructed machine learning model; To address the problem of poor classification and recognition of small samples in EEG motor imagery, the FBCapsNet model is proposed by integrating the features of DHDANet into the main

capsule layer of the capsule network. In addition, a dual classification recognition mechanism is introduced in the FBCapsNet model, which first learns the vector weights of different rhythm segment feature attributes through capsule networks, and then classifies based on the vector weights of each rhythm segment through Soft Attention. Qiao [14] et al. use wavelet analysis to extract EEG signals based on left and right motion imagination, effectively extracting event related synchronous and asynchronous information, and achieving the time-frequency information extracting from EEG signals. Zhang [15] et al. use a joint method of Convolutional Neural Networks (CNN) and Long-Short-Term Memory Networks (LSTM) to extract EEG signals, reducing network gradient dispersing and enhancing the stability of the entire network model. Sun [16] et al. design a graph self-attention network (EGSAN) which is based on EEG for learning low dimensional embedding vectors for EEG graphs, which can be adopted to classify features that are difficult to distinguish in motor imagery tasks.

Hauke et al. [17] propose a convolutional EEG signal recognition method which is based on nueral network, and experiments showed that this method has good recognition performance and strong robustness on the motor imagery EEG dataset. Zhang et al. [18] use long short-term memory networks to classify and recognize the motor imagery of left and right hand. This method combines time and frequency domain features of the motor imagery of left and right hand to achieve high recognition accuracy and strong generalization ability. Ge Rongxiang et al. [19] achieved high recognition accuracy in recognizing multiple types of motor imagery EEG signals using deep belief networks. Tang Zhichuan et al. [20] used convolutional neural networks to extract and classify features of imagined left hand and foot movement EEG signals, achieving a highest recognition rate of 93.22% on public datasets. Duan et al. [21] used principal component analysis and LDA for features extracting from class II motor imagery EEG signals, and classified them using multi-layer extreme learning machines, achieving a recognition accuracy of 94.2%. The above all demonstrate that using deep learning methods has great advantages in the classification of the motor imagery of brain computer interface.

With the continuous deepening of algorithms for analyzing EEG signals, it has been proven that deep learning-based methods have great feasibility in the field of EEG signal analysis.

3 Algorithm Model

Traditional EEG signal feature extraction is often constrained by the EEG signals complexity, and the process of extraction is relatively complicated, with low recognition accuracy and weak generalization ability. This article utilizes the characteristics of convolutional networks that can extract data features easily and that of LSTM which can extract temporal features easily, and also uses ConvLSTM networks to analyze EEG signals.

3.1 Wavelet packet transform

EEG signals are non-stationary signals that are suitable for time-frequency domain analysis. Currently, the most widely used transform is wavelet transform, but wavelet transform has poor frequency resolution in the high frequency range and good time resolution in the low frequency range. Wavelet packet transform is

able to not only decompose low-frequency components of EEG waves, but also decompose high-frequency ones. This paper uses wavelet packet transformation to extract the μ and β rhythms of EEG signals.

Assuming the filtered EEG signal is $x_i^j(t)$, $x_i^j(t) \in R^{N*n*m}$, where $x_i^j(t)$, $x_i^j(t) \in R^{N*n*m}$ and N refers to the total number of samples (total number of subjects in the experiment), n refers to the number of leads, and m is the number of sampling points.

The process of wavelet packet transform is the decomposition of the signal into multiple wavelet packets, and then perform wavelet transform on each wavelet packet to obtain wavelet coefficients for better understanding of signal characteristics.

Using wavelet packets to decompose the signal $x_i^j(t)$, the decomposition coefficients of wavelet of the first layer and k-th point are:

$$d_l^{2n}(k) = \sum_h g_0(h-2k) d_{l-1}^n(h) \quad (1)$$

$$d_l^{2n+1}(k) = \sum_h g_1(h-2k) d_{l-1}^n(h) \quad (2)$$

Among them, $g_0(h-2k)$ and $g_1(h-2k)$ are a pair of conjugate orthogonal filters, where h is the filter coefficient number and n are the wavelet packet subspace number.

After wavelet packet decomposition of the signal, select the wavelet packet subspaces of the rhythm (8-12Hz) and rhythm (18-26Hz) parts for EEG reconstruction. The reconstruction formula is as follows:

$$d_{l+1}^r(k) = \sum_h g_0(h-2k) d_l^{2r}(h) + \sum_h g_1(h-2k) d_l^{2r+1}(h) \quad (3)$$

3.2 ConvLSTM network

In LSTM, networks are used to capture spatiotemporal dependencies in datasets, and convolutional networks can be used to extract spatial features of data. ConvLSTM combines the characteristics of CNN and LSTM network models, which can not only obtain temporal dependencies, but also extract spatial features of data like convolutional layers, thus obtaining spatiotemporal features of signals. Signals of EEG are complex time series, and in the recognition of motor imagery in EEG signals, the motor imagery signals category is determined by the features of the EEG sequence. Although some useful information has a relatively long interval, it still needs to be retained. This article chooses to use ConvLSTM neural network for recognizing EEG signals of motor imagery. The basic structure of ConvLSTM is shown as Figure 1.

The difference between ConvLSTM and traditional LSTM is that ConvLSTM transforms the feedforward method in LSTM from Hadamard product to convolution operation. The ConvLSTM basic structure is shown above in the figure. ConvLSTM uses three gates for control, respectively the forget gate, input gate, and output gate. The door determines the way information passes through, protects and controls the cell status.

3.2.1 Forgetting Gate. In the gate structure of ConvLSTM, sigmoid function is used as an activating function. In the forgetting gate, both the information retained from the previous hidden state and the current input information are fed into the sigmoid function. This function output range is between [0,1] and this decides which information should be discarded or remained. The closer it is to 0, the more it should be discarded, and the closer it is to 1, the

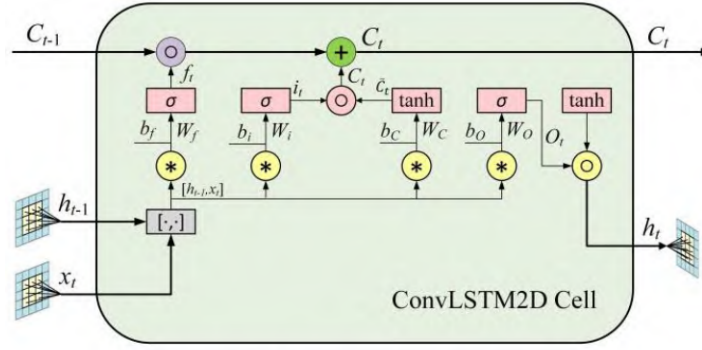


Figure 1: The basic structure of LSTM.

more it should be retained. The LSTM forget gate controls how much of the previous long-term state C_{t-1} will be output to the current long-term state C_t , and its calculating method is illustrated in formula (4), where h_t represents the short-term state of time t . h_{t-1} represents the short-term state of the previous layer. w_{xf} , w_{cf} represent for the input data information, and b_f refers to the weight matrix and bias of the forget gate, respectively.

$$f_t = \sigma(w_{xf} * x_t + w_{hf} * H_{t-1} + w_{cf} * C_{t-1} + b_f) \quad (4)$$

•represents inner product operation, * represents convolution operation.

3.2.2 Input gate. The input gate controls how much information in the input x_t at the current moment will be saved to the current long-term state C_t . The previous layer hidden state and the current information of input will be passed to the sigmoid function, and the value will be adjusted to between 0-1 to determine which information to update. 0 represents unimportant, and 1 represents important. In this model, it transfers the hidden state of the previous layer and the current information of input to the tanh function, then creates a new candidate value vector, and finally multiplies the sigmoid value of output with the tanh value of output. The sigmoid output value will determine which information in the tanh output value is important and needs to be retained. The calculation formula is as follows, where w_{xi} , which, w_{hi} , w_{xc} , w_{hc} , and b_i are the biases and weights of the input gates, respectively.

$$i_t = \sigma(w_{xi} * x_t + w_{hi} * H_{t-1} + w_{ci} * C_{t-1} + b_i) \quad (5)$$

$$\tilde{c}_t = \tanh(W_{xc} * x_t + W_{hc} * h_{t-1} + b_c) \quad (6)$$

By using input gates and forget gates, the long-term cell state C_{t-1} can be fused with the newly generated cell state $\tilde{c}_t$ according to the following equation to obtain the long-term state C_t at the current time. When $\tilde{c}_t$ is the newly generated state of cell corresponding to the input at the current time, the calculation method is as follows.

$$C_t = f_t * C_{t-1} + i_t * \tilde{c}_t \quad (7)$$

3.2.3 Output gate. The output gate of LSTM is to control how much of the current long-term state C_t will be output to the current output value h_t . The calculation method is as follows. Among them, w_{xo} , w_{ho} , w_{co} , and b_o are the biases and weights of the output,

respectively. The final output h_t of the network is output by C_t through the output gate.

Output of the model is determined by the output gate. Firstly, an initial output is obtained through the sigmoid layer. The information in previous hidden state and the current input are passed to the sigmoid function, and then the newly obtained cell state is fed to the tanh function. The tanh function is used to scale the value to -1 to 1, and then the output obtained from the sigmoid is multiplied pairwise to obtain the output of the model.

$$O_t = \sigma(w_{xo} * x_t + w_{ho} * h_{t-1} + w_{co} * c_t + b_o) \quad (8)$$

$$h_t = O_t * \tanh(C_t) \quad (9)$$

The output of sigmoid function does not contain the information which is learned in previous time, while the tanh function compresses the information which is learned in the previous time, taking the role of stabilizing the numerical value.

3.2.4 A Classification Model for Motion Imagination Signals Based on ConvLSTM. The model for classifying motor imagery signals in this article is shown in the following figure. Firstly, preprocess the original signal to obtain raw EEG signals related to motor imagery. Normalize signal data to eliminate individual differences. Then, the ConvLSTM model is used to generate EEG feature sequences, and the model is trained using a long short-term memory recurrent network model. Finally, output the classification results. It is divided into the following four steps.

Step 1: Use wavelet packet transform to extract signals related to the frequency of motion imagination signals.

Step 2: Use convolution operation to extract sequence features of EEG signals.

Step 3: Input the generated EEG feature sequence into the LSTM network for model training.

Step 4: Output classification results.

4 Experiment Simulation

In the experiment, Python language is used for data preprocessing and model construction. During the experiment, a GPU server is adopted to train and test the prediction model, and Pytorch is chosen as the model framework. The specific experimental environment is illustrated in the Table 1 below.

Table 1: The information of experimental environment.

ENVIRONMENT	VERSION
operating system	Ubuntu 20.04
storage	64G
CPU version	Xeon Gold 5218R
GPU version	NVIDIA A5000
CUDA version	12.1
Pytorch version	1.12
Python version	Python3.8

4.1 The Experimental Data

The dataset adopted in the experiment is the EEG Motor Movement/Imagery Dataset (eegmmidb), which includes four tasks: separately opening and closing the hand, imagining opening and closing the left and right hands, opening and closing both hands and feet, and imagining opening and closing both hands and feet. The EEG data of 64 channels in the human brain is included.

In this experiment, we chose to classify only the left and right hands, so we chose Task 2. The tester's states include T0: resting state, T1: imagining the left hand releasing/gripping, T2: imagining the right hand releasing/gripping.

Each exercise imagination experiment lasts for 4 seconds, with a sampling rate of 160 Hz, which means there are 640 sampling points in a single imagination experiment. Use the data from the top 80 subjects (S01-S80) containing Task 2 as the dataset used in this experiment. In the experiment, only the movements of the left and right hands are identified. We took tasks 4, 8, and 12 from 80 datasets in the database S001-S080 as the experimental dataset. 80% of it is used as the training set for learning, constructing and running imagination classification model. 10% is used as the validating set to validate the model and construct the optimal network structure, while 10% is used as the testing set to evaluate the recognition accuracy of the model.

4.2 Data preprocessing

1. Artifact removal

Artifacts are noise that needs to be eliminated. The noise is mainly slow drift and power line noise.

Low frequency drift is usually below 1Hz. Therefore, we set up a filter to remove drift below 1Hz. Power source noise is the noise generated by the power grid. It consists of peaks of 60Hz. Some peaks may also occur at harmonic frequencies, which are integer multiples of the power line frequency, such as 120Hz, 180Hz, etc.

We will remove 220V AC 60Hz power supply noise as well as its harmonics, such as 100Hz, 150Hz, etc. Since our signal is 80Hz (according to Nyquist's theorem, it is $160\text{Hz}/2$), we do not need to run harmonics, but only need to intercept the 60Hz signal. To eliminate slow drift, high pass filtering is used. The processed image is as follows, and we can see that the 60Hz noise has been removed.

2. Low pass filtering

The signals to be analyzed in this article are μ rhythm (8-12Hz) and β rhythm (18-26Hz). Therefore, a 50Hz low-pass filter is applied to the original EEG data, so that we can use wavelet packet transform to reconstruct and extract signals of Mu rhythm and Beta

rhythm from the data. After low-pass filtering, the signal is shown in the following figure.

3. ICA

The idea of programming of ICA in MNE is firstly building an ICA object (which is corresponding to constructing an ICA analyzer) and then using this ICA analyzer to analyze the EEG data (through ICA object means).

Since ICA is not effective for low frequency data, here ICA and artifact components are removed based on high-pass 1Hz data and then applied to high-pass 0.1Hz data.

Eye movements, muscle, blinking, heart and line noise can seriously contaminate the activity of EEG, which would cause serious problems in EEG analysis and interpretation. Researchers propose applying the ICA method to multi-channel EEG, recording and removing different kinds of artifacts from EEG records by obviating the impacts of human factors on scalp sensors. The results of the research indicate that ICA can effectively detect, separate, and eliminate different kinds of artifacts in EEG records, and its results are more advantageous compared to those obtained through regression based or PCA based means.

The EEG signal waveform after the processing above is shown in Figure 2. From the graph, it can be seen that frequencies below 1Hz and data at 60Hz have been removed. After filtering, data above 50Hz has all been removed.

4. Wavelet packet transform

This article uses wavelet packet technology to extract mu rhythms and beta rhythms, and calculates their energy mean as features. Based on the frequency band of the signal to be extracted, select the corresponding wavelet coefficients for signal reconstruction, and the corresponding signal can be extracted.

This paper adopts db4 as the wavelet basis function for 3-level wavelet decomposition, forming 8 nodes. We use a 50Hz low-frequency filter, and the frequency range after wavelet packet decomposition is shown in the figure. Due to the frequency range of Mu rhythm and Beta rhythm being 8-12Hz and 18-26Hz respectively, we use the second, third, fourth, and fifth frequency bands and superimpose them to extract the desired μ rhythm and β rhythm signals.

Figure 3 shows the energy of the data in each frequency band after wavelet packet decomposition. It can be seen from the figure that the energy of the EEG signal is mainly concentrated in the Delta frequency band.

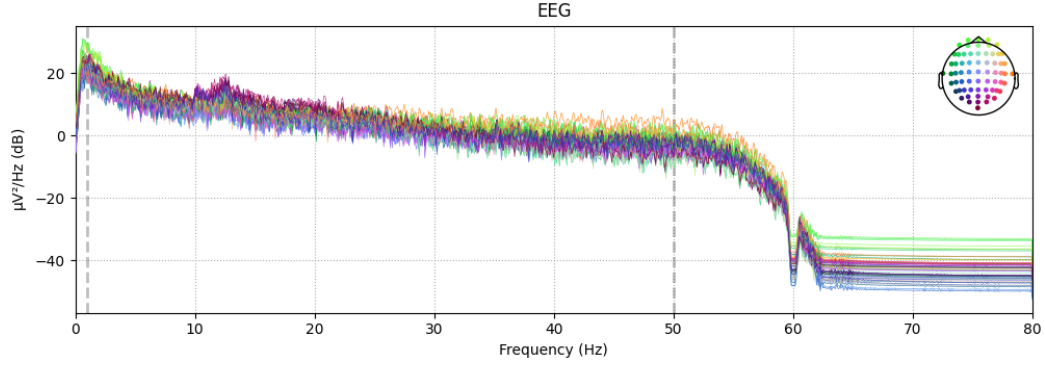


Figure 2: EEG data after preprocessing.

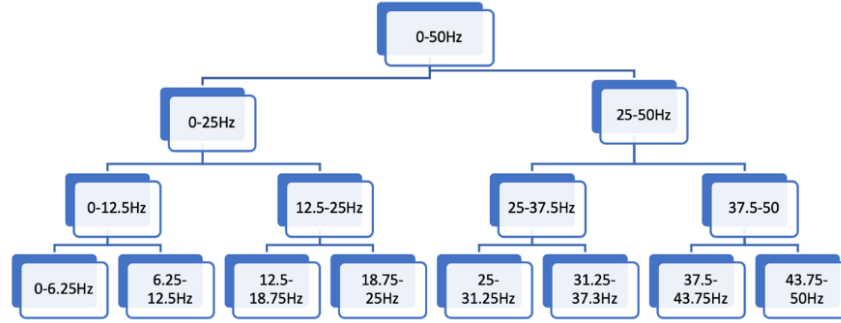


Figure 3: Wavelet packet number.

4.3 Experimental Results and Analysis

The commonly used classification performance indicators in classification problems include accuracy, precision, recall rate, F1 score, etc.

1. Accuracy means the percentage of correctly predicted results in the total sample, which is calculated by the following formula.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \quad (10)$$

2. Accuracy means the proportion of actual correct samples of that predicted by the model as positive examples. Which is calculated by the following formula.

$$Precision = \frac{TP}{TP + FP} \times 100\% \quad (11)$$

3. Recall rate means the percentage of samples that are truly positive and predicted by the model to be positive. The higher the recall rate, the more positive samples are correctly predicted, and the better the predictive performance of the model, which is calculated by the following formula.

$$Recall = \frac{TP}{TP + FN} \times 100\% \quad (12)$$

4. F-Measure, also known as F1 score, comprehensively considers the performance of classification models in terms of recall and

accuracy. The calculation formula is as follows.

$$F1 = \frac{2 \times Recall \times Precision}{Recall + Precision} \quad (13)$$

True Positive (TP) represents the number of classifications that accurately predict positive cases as positive cases; True Negative (TN) represents the number of categories that correctly predict negative cases as negative cases; False Positive (FP) represents the number of classifications that incorrectly predicted negative examples as positive examples; False Negative (FN) represents the number of categories that incorrectly predicted positive cases as negative cases.

1. The confusion matrix before and after using wavelet transform is shown in the following Figures 4 and 5

2. The comparison of various performance indicators before and after using wavelet transform is shown in Table 2.

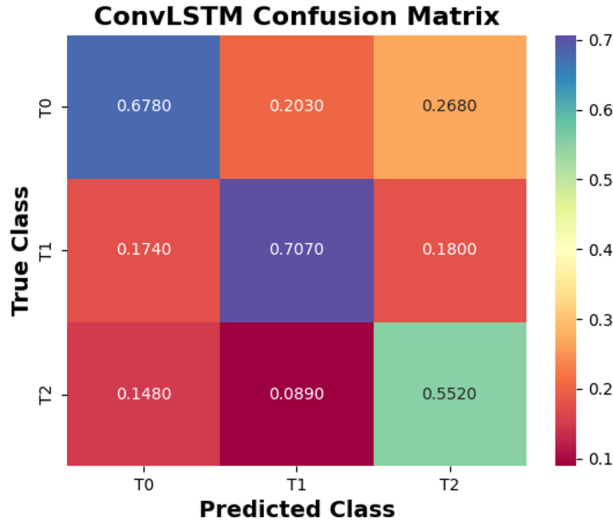
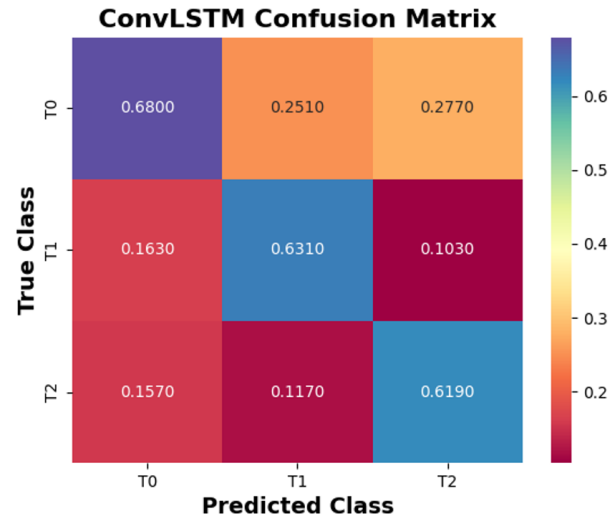
According to Table 2, it is seen that the accuracy, recall, and F1 Score have all improved after using wavelet transform, which means it is feasible to use wavelet packet transform for feature extracting of EEG signals.

5 conclusions

This article is based on the spatiotemporal characteristics of motor imagery EEG signals, combined with the advantages of convolutional operations and LSTM, and uses the ConvLSTM network

Table 2: Parameter comparison.

	WaveletPackage-ConvLSTM	ConvLSTM
Precision	0.6537	0.6480
Recall	0.6300	0.6157
F1-Score	0.6354	0.6206

**Figure 4: Confusion matrix without using wavelet transform.****Figure 5: Extracting Feature Confusion Matrix Using Wavelet Packets.**

model as the classification method for EEG signals. Firstly, the frequency features of the EEG signal are extracted using wavelet packet transform. Secondly, the EEG signal spatial features are

extracted through convolution operation. Then, the temporal features are extracted through LSTM operation. Finally, the extracted features are classified through a classification layer. This method demonstrates that the algorithm is feasible. Also, compared to other methods of EEG recognition, this method is more efficient in catching local features and establishing time-sequence relationship.

In this experiment, we only validated the role of wavelet packet transform in this method. The structure of ConvLSTM network was not verified, and deep learning experiments were not conducted. Additionally, only one convolutional layer was used for feature extraction in the network, so the accuracy of the algorithm is not very high. In future experiments, we will further investigate the impact of network depth on the performance of the algorithm.

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